Practical 2

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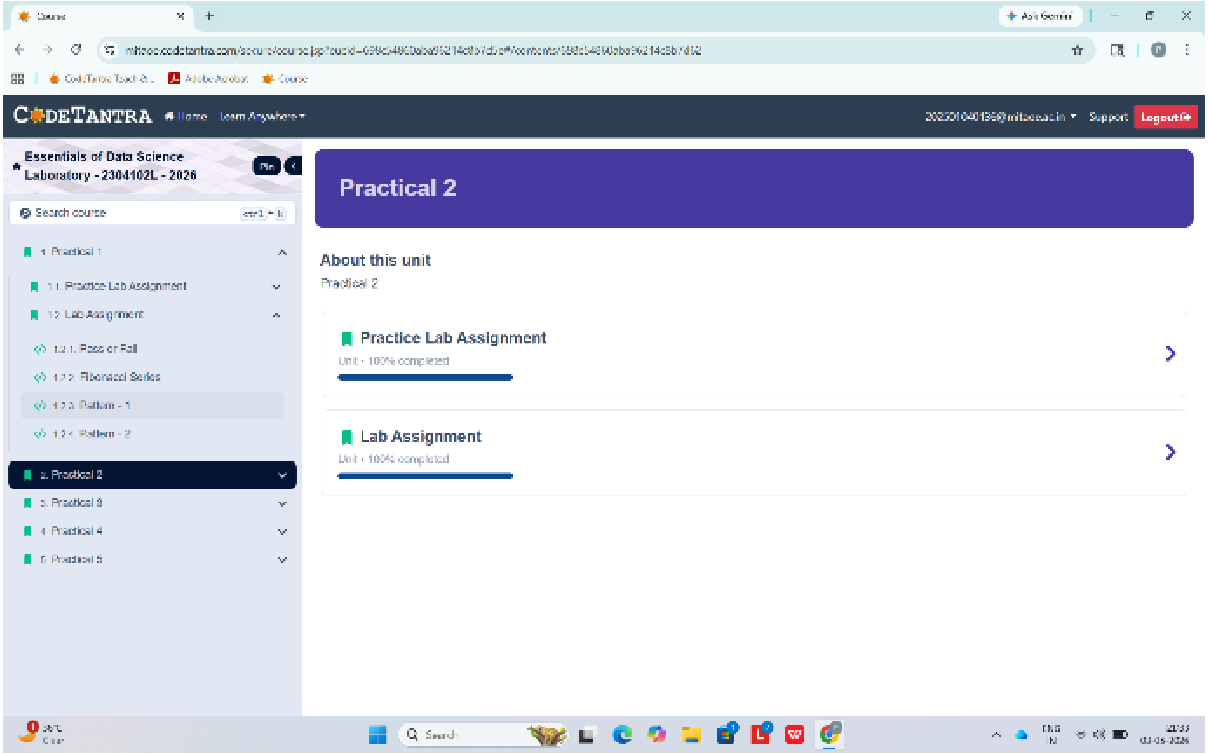
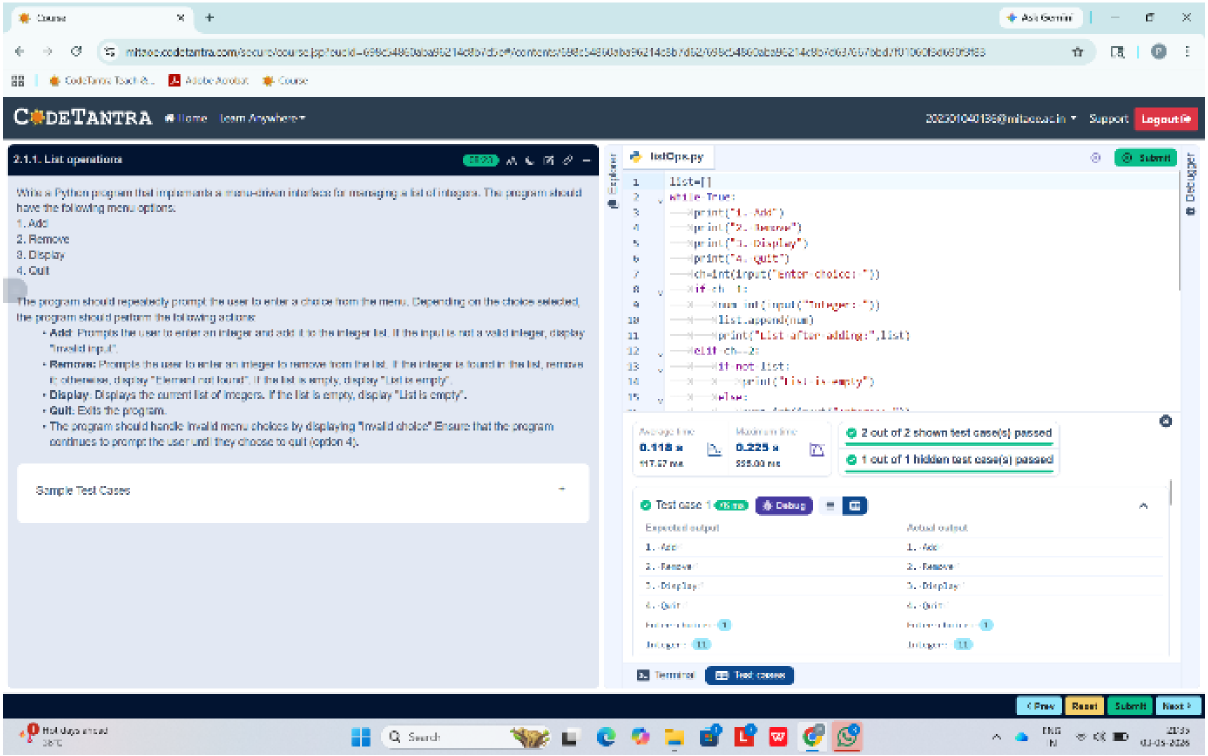
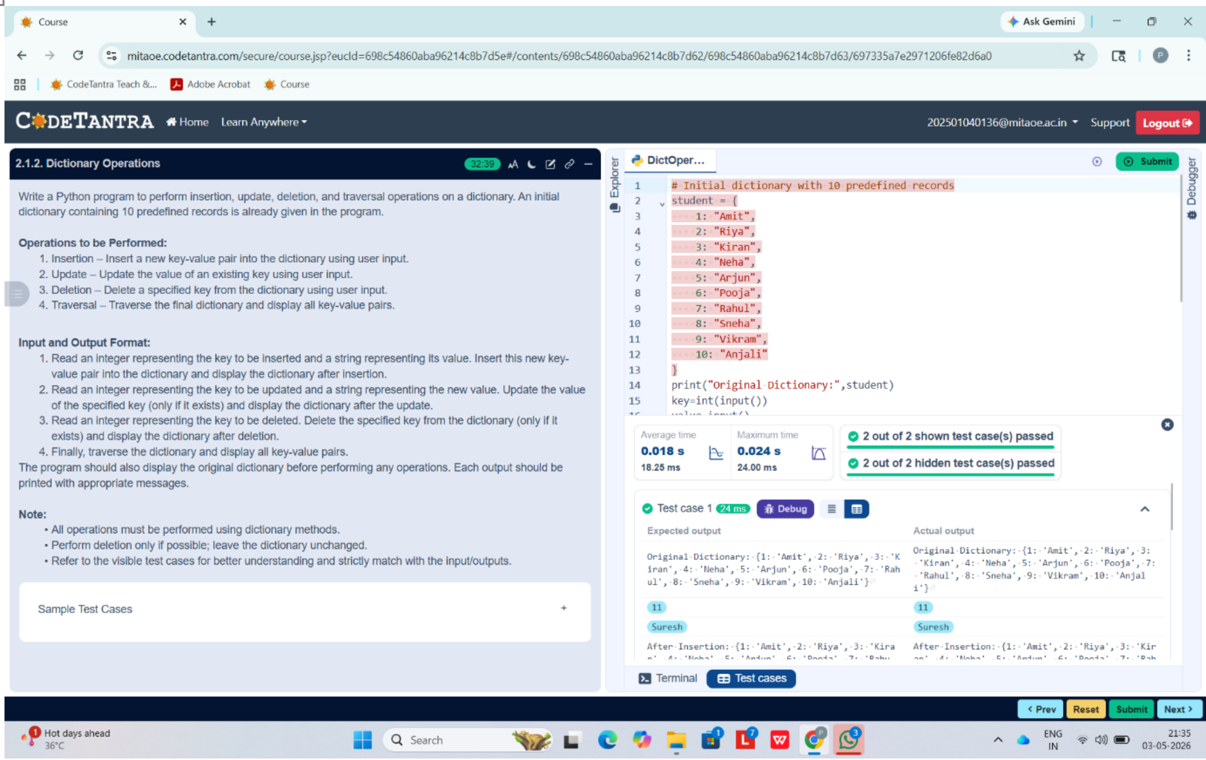






2.1.2. Dictionary Operations

Write a Python program to perform insertion, update, deletion, and traversal operations on a dictionary. An initial dictionary containing 10 predefined records is already given in the program.

Operations to be Performed:

1. Insertion – Insert a new key-value pair into the dictionary using user input.
2. Update – Update the value of an existing key using user input.
3. Deletion – Delete a specified key from the dictionary using user input.
4. Traversal – Traverse the final dictionary and display all key-value pairs.

Input and Output Format:

1. Read an integer representing the key to be inserted and a string representing its value. Insert this new key-value pair into the dictionary and display the dictionary after insertion.
2. Read an integer representing the key to be updated and a string representing the new value. Update the value of the specified key (only if it exists) and display the dictionary after the update.
3. Read an integer representing the key to be deleted. Delete the specified key from the dictionary (only if it exists) and display the dictionary after deletion.
4. Finally, traverse the final dictionary and display all key-value pairs.

The program should also display the original dictionary before performing any operations. Each output should be printed with appropriate messages.

Note:

- All operations must be performed using dictionary methods.
- Perform deletion only if possible; leave the dictionary unchanged.
- Refer to the visible test cases for better understanding and strictly match with the input/outputs.

Sample Test Cases

```python
# Initial dictionary with 10 predefined records
student = {
    1: "Amit",
    2: "Riya",
    3: "Kiran",
    4: "Neha",
    5: "Arjun",
    6: "Pooja",
    7: "Rahul",
    8: "Sneha",
    9: "Vikram",
    10: "Anjali"
}
print("Original Dictionary:",student)
key=int(input())
```

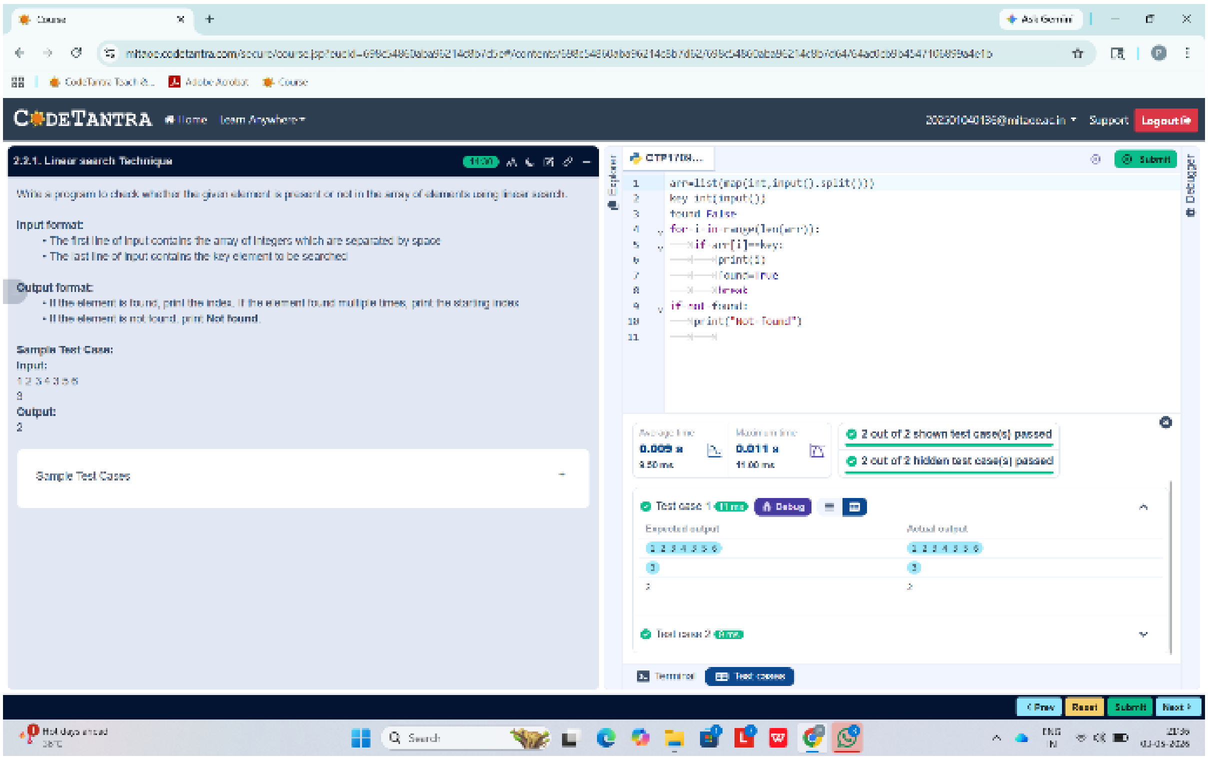
Average time 0.018 s 18.25 ms | Maximum time 0.024 s 24.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 24 ms

Expected output	Actual output
Original Dictionary: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali'}	Original Dictionary: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjal i'}
11	11
Suresh	Suresh
After Insertion: {1: 'Amit', 2: 'Riya', 3: 'Kira	After Insertion: {1: 'Amit', 2: 'Riya', 3: 'Kir



2.2.1. Linear search Technique

Write a program to check whether the given element is present or not in the array of elements using linear search.

Input format:

- The first line of input contains the array of integers which are separated by space
- The last line of input contains the key element to be searched

Output format:

- If the element is found, print the index. If the element found multiple times, print the starting index
- If the element is not found, print **Not found**.

Sample Test Case:

Input:

1 2 3 4 3 5 6

3

Output:

2

Sample Test Cases

```python
arr=list(map(int,input().split()))
key=int(input())
found=False
for i in range(len(arr)):
    if arr[i]==key:
        print(i)
        found=True
        break
if not found:
    print("Not found")
```

Average time 0.009 s 9.50 ms | Maximum time 0.011 s 11.00 ms

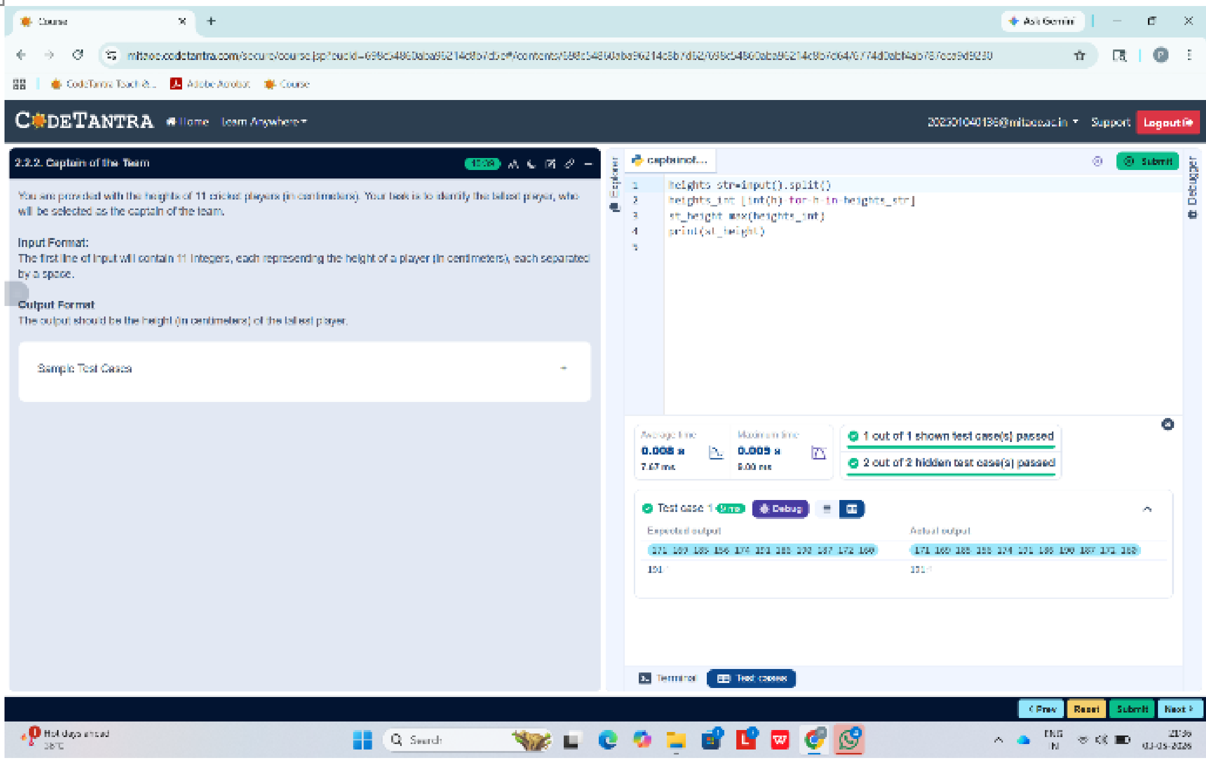
2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 11 ms

Expected output	Actual output
1 2 3 4 3 5 6	1 2 3 4 3 5 6
3	3
2	2

Test case 2


CODETANTRA
Home
learn Anywhere
Support
Logout
2.2.2. Captain of the Team
You are provided with the heights of 11 cricket players (in centimeters). Your task is to identify the tallest player, who will be selected as the captain of the team.
Input Format:
The first line of input will contain 11 integers, each representing the height of a player (in centimeters), each separated by a space.
Output Format
The output should be the height (in centimeters) of the tallest player.
Sample Test Cases
Submit
heights_str=input().split()
heights_int [int(h) for h in heights_str]
st_height max(heights_int)
print(st_height)
1 out of 1 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1
Debug
Expected output
Actual output
Terminal
Test cases
Prev
Reset
Submit
Next
Search